Health Monitoring API reference

Version 1.1 · 13 Oct 2026 · Shubham Sinha

The module exposes 13 REST endpoints under `/api/v1`. They follow the app design: one editable health log per patient per day, with a status for each reading and for the day.

Version 1.1 answers the review of 13 Oct 2026. Sections marked NEW or CHANGED are listed in Changes in this version. Clinical values that are still placeholders are listed in Open questions.

Conventions

Item	Rule
Base URL	`/api/v1`
Auth header	`Authorization: Bearer <Firebase ID token>`
Request ID NEW	Optional `X-Request-Id` header. The server creates one if missing, returns it on every response and stores it in audit records.
Day CHANGED	The patient's local calendar day, using the IANA time zone in their profile (for example `Asia/Kolkata`). See Day and time zone
Times	ISO 8601 with offset, for example `2026-10-09T07:05:00+05:30`
Dates	`YYYY-MM-DD`, a local calendar day of the patient
IDs	MongoDB ObjectId strings
Pagination	`?limit=20&cursor=<nextCursor>`. Lists return `{ data, nextCursor }`. Maximum limit is 100
Errors CHANGED	One format with a stable machine-readable `code`. See Errors
Docs	Swagger UI at `/api/docs`, generated from the NestJS decorators

Endpoints

Method	Route	Role	Purpose
GET	`/vital-types`	Any	Parameters, units and valid ranges, to build the form
GET	`/me/vitals/today`	Patient	Today's log with status, to pre-fill the form and Health tab
PUT	`/me/vitals/today`	Patient	Save or update some or all of today's readings
GET	`/me/vitals/logs`	Patient	Past daily logs
GET	`/me/vitals/trends`	Patient	Daily trend series for one parameter
GET	`/doctor/patients/:patientId/vitals/today`	Doctor	A patient's log for today

Method	Route	Role	Purpose
GET	`/doctor/patients/:patientId/vitals/logs`	Doctor	A patient's past logs
GET	`/doctor/patients/:patientId/vitals/trends`	Doctor	A patient's trends
GET	`/doctor/alerts`	Doctor	Alerts across the doctor's patients
PATCH	`/doctor/alerts/:alertId`	Doctor	Acknowledge an alert
GET	`/admin/alert-thresholds`	Admin	List alert limits
PUT	`/admin/alert-thresholds/:type`	Admin	Set an alert limit, globally or for one patient
PUT	`/admin/vital-types/:type`	Admin	Change a parameter's unit or valid range

There are no POST or DELETE routes for readings. A day's log is created on its first save and is never deleted (see Reading lifecycle).

Parameters CHANGED

All 13 parameters come from the SRS (Daily Health Monitoring). The current app design shows 9 of them, so walking duration, sleep duration, nutrition intake and water intake need to be added to the form.

type	Parameter	value	Unit	Valid range (`minValid` – `maxValid`)	Fields counted
`BP`	Blood pressure (upper / lower)	`{ systolic, diastolic }`	mmHg	systolic 60–260, diastolic 30–160, each checked on its own	2
`PULSE`	Pulse rate	`{ num }`	bpm	30–220	1
`SPO2`	Oxygen saturation (SpO2)	`{ num }`	%	50–100	1
`TEMP`	Temperature	`{ num }`	°C	32.0–43.0	1
`WEIGHT`	Weight	`{ num }`	kg	20–250	1
`SUGAR`	Blood sugar (optional)	`{ num }`	mg/dL	20–600	1
`PAIN`	Pain score	`{ num }`, whole number	/10	0–10	1
`WALK_MIN`	Walking duration	`{ num }`, whole number	minutes	0–1440	1
`STEPS`	Step count (manual)	`{ num }`, whole number	steps	0–100000	1
`SLEEP_HRS`	Sleep duration	`{ num }`, 0.5 steps	hours	0–24	1
`NUTRITION`	Nutrition intake	`{ text }`		1–500 characters	1
`WATER_ML`	Water intake	`{ num }`, whole number	ml	0–10000	1

type	Parameter	value	Unit	Valid range (`minValid` – `maxValid`)	Fields counted
`MOOD`	Mood assessment	`{ num }` , 1–5, returned with a `label`	level	1–5	1

That gives 14 fields on the form. Mood level 4 is labelled "Good" in the design, and the other four labels are pending from the frontend team.

Validation is not alerting. The valid range only rejects typing mistakes and impossible values, with a 400. A value inside the range is always saved. Whether it is clinically worrying is decided separately by the alert thresholds. The ranges above are initial data-entry limits that an admin can change. They are not clinical limits.

Versioning. Every parameter has a `version` , starting at 1. Any admin change to its unit or range adds 1. Each saved reading stores the `unit` and `typeVersion` it was saved under, so old values stay readable after a change. A parameter's value shape (`valueKind`) never changes. A new shape needs a new type code.

Status

Every log carries a status, so the app can show "Not yet" and "9 of 14 logged" without working it out itself.

Field	Values	Meaning
`readings.<type>.status`	`LOGGED` , `NOT_YET`	Whether that parameter has a value today
`filledCount`	0–14	Fields filled today. BP upper and lower count separately
`totalCount`	14	Fields of the parameters where `onForm` is true. 14 when all are on the form
`dayStatus`	`NOT_STARTED` , `IN_PROGRESS` , `COMPLETE`	0 filled, some filled, or all fields filled

Day and time zone NEW

- All times travel as ISO 8601 with an offset. The server stores them in UTC.
- "Today" means the current calendar date in the patient's profile time zone, worked out by the server when the request arrives.
- Each daily log stores its `date` and the `timeZone` used to work it out. A later change to the profile time zone only affects future days. Past logs keep their date.
- Trend labels and the `from` / `to` filters use these local dates.

GET /vital-types CHANGED

The app builds the form from this list. Response `200` :

```
{ "data": [
  { "type": "BP", "label": "Blood pressure", "unit": "mmHg", "valueKind": "pair",
    "fields": { "systolic":  { "minValid": 60, "maxValid": 260 },
                "diastolic": { "minValid": 30, "maxValid": 160 } },
    "isOptional": false, "onForm": true, "version": 1 },
  { "type": "SPO2", "label": "SpO2", "unit": "%", "valueKind": "numeric",
    "minValid": 50, "maxValid": 100, "isOptional": false, "onForm": true, "version": 1 },
  { "type": "MOOD", "label": "Mood", "unit": "level", "valueKind": "scale",
    "minValid": 1, "maxValid": 5, "isOptional": false, "onForm": true, "version": 1 }
] }
```

`valueKind` is one of `numeric`, `pair` (BP only), `scale` or `text`. For `text`, `minValid` and `maxValid` are character counts. Errors: `401`.

GET /me/vitals/today

Returns today's log. If nothing is logged yet, every parameter is `NOT_YET` and `dayStatus` is `NOT_STARTED`. Response `200`:

```
{ "date": "2026-10-09", "timeZone": "Asia/Kolkata", "dayStatus": "IN_PROGRESS",
  "filledCount": 9, "totalCount": 14,
  "readings": {
    "BP":    { "value": { "systolic": 128, "diastolic": 82 }, "unit": "mmHg",
              "status": "LOGGED", "loggedAt": "2026-10-09T07:05:00+05:30" },
    "PULSE": { "value": { "num": 76 }, "unit": "bpm",
              "status": "LOGGED", "loggedAt": "2026-10-09T07:05:00+05:30" },
    "SUGAR": { "value": null, "unit": "mg/dL",
              "status": "NOT_YET", "loggedAt": null },
    "PAIN":  { "value": { "num": 7 }, "unit": "/10",
              "status": "LOGGED", "loggedAt": "2026-10-09T15:03:00+05:30" },
    "MOOD":  { "value": { "num": 4 }, "label": "Good",
              "status": "LOGGED", "loggedAt": "2026-10-09T15:03:00+05:30" }
  } }
```

The example shows 5 of the 13 parameters. The real response always includes all 13. Errors: `401`, `403`.

PUT /me/vitals/today CHANGED

Saves or updates today's log. Send only the fields that changed. Fields you don't send stay as they are, which matches "Fill in what you can. You can come back later."

Request:

```
{ "date": "2026-10-09",
  "readings": {
    "WEIGHT": { "num": 56 },
    "STEPS":  { "num": 5000 },
    "PAIN":   { "num": 7 },
    "MOOD":   { "num": 4 }
  } }
```

Rules

- `date` is required. It is the day the form was opened for, and it must equal the patient's today. If the day has changed, for example a retry after midnight, the server returns `409 DAY_CHANGED` and saves nothing. The app then reloads today's log.
- At least one reading must be sent. Unknown type codes are rejected.
- Each value must match its type's shape and be within its valid range. For `BP`, both fields are sent together, each is checked against its own range, and systolic must be greater than diastolic.
- Sending `null` clears a value, and its status goes back to `NOT_YET`.
- The server sets `loggedAt` for each changed field. The app doesn't send times.

Duplicates and retries NEW

- A log holds one value per parameter per day. Saving a parameter again replaces its value. It never adds a second reading.
- A sent value equal to the stored value counts as unchanged. It keeps its `loggedAt` and creates no audit entry, alert or event.
- So the request is idempotent. Re-sending the same body after a timeout gives the same log and no duplicate alerts. No `Idempotency-Key` is needed.
- Two devices saving at once are merged field by field. For the same field, the last save wins, and both saves are in the audit log.

Response `200`: the full log, as in `GET /me/vitals/today`, plus any alerts raised or updated by this save:

```
{ "date": "2026-10-09", "timeZone": "Asia/Kolkata", "dayStatus": "IN_PROGRESS",
  "filledCount": 9, "totalCount": 14,
  "readings": { "…": "as in GET /me/vitals/today" },
  "alerts": [
    { "id": "6705a1c2e4b0f1a2b3c4d5f0", "type": "PAIN", "field": null,
      "rule": "ABOVE_HIGH", "threshold": 6, "severity": "MEDIUM", "status": "OPEN" } ] }
```

Errors: `400 VALIDATION_FAILED` invalid or empty body, `401` missing or expired token, `403` caller is not a patient, `409 DAY_CHANGED`, `429` too many requests.

Save flow: what happens when NEW

Steps 1–7 finish before the response is sent. Step 8 runs afterwards.

1. Verify the Firebase token and the patient role.
2. Take the patient ID from the token, never from the request.
3. Validate the body shape, type codes, value shapes and valid ranges.
4. Check that `date` is the patient's today.
5. Drop fields whose value is unchanged.
6. In one MongoDB transaction: update the daily log, recount the status, create or update alerts, write audit entries, and write the events to an outbox collection.
7. Commit, then return the log and alerts.

8. After the commit, a publisher sends the outbox events. The Timeline and Notifications modules handle them on their own.

If step 6 fails, nothing is saved. A failure in step 8 never undoes a saved reading. The event stays in the outbox and is retried.

Reading lifecycle NEW

- **Today:** the patient can add, change or clear (`null`) any value until the end of their local day.
- **Past days:** read-only. There is no route to edit or delete a past log in version 1. Whether patients may correct yesterday is an open question.
- **Delete:** logs and readings are never deleted. Clearing a value is the only removal, and it is audited.
- **History:** the old and new value of every change are kept in the audit log. Doctors always see the current value.
- **Alerts after an edit:** see Alert lifecycle. Changing or clearing a value never deletes or closes an alert.

GET /me/vitals/logs

Returns past daily logs, newest first, in the same shape as today's log.

Query parameter	Required	Default	Notes
`from`	No	30 days ago	Date, for example `2026-09-10`
`to`	No	Today	Date. `from` must not be after `to`
`limit`	No	20	1–100
`cursor`	No		`nextCursor` from the previous page

Response `200` : `{ "data": [ <daily log>, … ], "nextCursor": "…" }` . Days with no saves are left out. Errors: `400` , `401` , `403` .

GET /me/vitals/trends CHANGED

Returns one value per day for one parameter, shaped for Chart.js.

Query parameter	Required	Default	Notes
`type`	Yes		Any parameter except `NUTRITION`
`period`	No	`30d`	`7d` , `30d` or `90d`

```
{ "type": "PULSE", "unit": "bpm", "period": "7d",
  "labels": ["2026-10-03", "2026-10-04", "2026-10-05", "2026-10-06",
             "2026-10-07", "2026-10-08", "2026-10-09"],
  "values": [78, 80, null, 77, 75, 79, 76],
  "thresholds": { "low": 50, "high": 110 } }
```

How each day's value is chosen

- A log holds one value per parameter per day, so the trend value is that day's current saved value. No averaging is done, and this holds for every parameter, including water, steps, walking and sleep.
- The period is the last N local days, ending with today.
- Days with no value are `null`, so the chart shows a gap.
- For `BP`, the response has `systolic` and `diastolic` arrays instead of `values`, and `thresholds` has one entry per field.
- `thresholds` are the limits in effect today for this patient: the patient override if one exists, else the global default. It is `null` when no limit is set.

Errors: `400` unknown type or `NUTRITION`, `401`, `403`.

Doctor endpoints

The three `/doctor/patients/:patientId/vitals…` routes take the same query parameters and return the same shapes as the matching `/me/vitals…` routes. They add two rules:

- `PatientAccessGuard` returns `404` when the patient isn't linked to the calling doctor, so the doctor can't tell whether that patient exists.
- Every successful call is recorded in the audit log: doctor, patient, route, request ID and time. The record is written after the access check passes. If it can't be written, the call fails with `500` rather than returning data unaudited.

GET /doctor/alerts

Query parameter	Required	Default	Notes
`status`	No	`OPEN`	`OPEN` or `ACKNOWLEDGED`
`patientId`	No	All linked patients	Filter to one patient. `404` if not linked
`limit`, `cursor`	No	20	As in logs

Response `200`, newest first:

```
{ "data": [
  { "id": "6705a1c2e4b0f1a2b3c4d5f0",
    "patient": { "id": "6701f0a9e4b0f1a2b3c4d001", "name": "Rajesh Iyer" },
    "date": "2026-10-09", "type": "PAIN", "field": null,
    "value": { "num": 7 }, "currentValue": { "num": 7 },
    "rule": "ABOVE_HIGH", "threshold": 6,
    "thresholdId": "6702c3d4e4b0f1a2b3c4d101", "thresholdVersion": 1,
    "severity": "MEDIUM", "status": "OPEN",
    "createdAt": "2026-10-09T15:03:02+05:30", "updatedAt": "2026-10-09T15:03:02+05:30",
    "acknowledgedBy": null, "acknowledgedAt": null, "note": null }
], "nextCursor": null }
```

`value` is the reading that crossed the limit. `currentValue` is that parameter's value in the log now, which may differ after the patient edits it.

PATCH /doctor/alerts/:alertId CHANGED

Request: `{ "status": "ACKNOWLEDGED", "note": "Called patient, pain settled after rest" }`. The note is optional, up to 500 characters.

Response `200`: the alert, with `acknowledgedBy` (doctor ID) and `acknowledgedAt` filled in. Errors: `400` any other status value, `404` alert not found or its patient isn't linked to this doctor, `409 ALERT_ALREADY_ACKNOWLEDGED`.

Alert lifecycle NEW

Question	Rule
When is an alert raised?	When a saved value is below `low` or above `high` of the limit in effect for that patient (patient override, else global). No limit set means no alert.
States	`OPEN` → `ACKNOWLEDGED`. Acknowledged is final: alerts never reopen.
De-duplication	At most one alert per patient, parameter, field and day. If a later save that day crosses the limit again, the same alert is updated (`value`, `severity` if higher, `updatedAt`). A new alert is raised only when the earlier one is already acknowledged.
Value edited back into range, or cleared	The alert stays `OPEN` until a doctor acknowledges it. Only `currentValue` changes, so the doctor sees both values.
Who can acknowledge	Any doctor linked to the patient. Patients and admins cannot.
Note	Set once when acknowledging. It cannot be edited afterwards.
Doctor notification	One `vitals.alert.raised` event when an alert is created, and again only if an update raises its severity.
Traceability	Each alert stores the threshold ID and version it was raised under. Later threshold changes don't recompute old alerts.

Admin endpoints

GET /admin/alert-thresholds CHANGED

Optional filters: `type`, `patientId`. Response `200`: `{ "data": [ <threshold>, … ], "nextCursor": … }`, current versions only.

PUT /admin/alert-thresholds/:type CHANGED

```
{ "field": null, "low": null, "high": 6, "severity": "MEDIUM", "patientId": null }
```

- If `patientId` is `null`, the rule is the global default. If it's set, the rule overrides the default for that patient.
- For `BP`, `field` is required: `systolic` or `diastolic`, each with its own limit. For other types it must be `null`.
- At least one of `low` or `high` is required, `low` must be less than `high`, and both must be inside the parameter's valid range.
- `severity` is `LOW`, `MEDIUM` or `HIGH`. Not allowed for `NUTRITION`.

- Each save creates a new version with `effectiveFrom` set to now. Earlier versions are kept, not overwritten. It applies to saves from then on.

Response `200`:

```
{ "id": "6702c3d4e4b0f1a2b3c4d101", "type": "PAIN", "field": null,
  "low": null, "high": 6, "severity": "MEDIUM", "patientId": null,
  "version": 2, "effectiveFrom": "2026-10-13T10:00:00+05:30", "updatedBy": "6700aa01e4b0f1a2b3c4d900" }
```

PUT /admin/vital-types/:type CHANGED

Request: any of `unit`, `label`, `minValid`, `maxValid` (or `fields` for `BP`), `isOptional`, `onForm`. `type` and `valueKind` cannot change. Response `200`: the updated parameter with its `version` increased by 1. Values already logged are not re-checked and keep the unit they were saved with.

Admin errors: `400`, `401`, `403` caller is not an admin, `404` unknown type or patient.

Events NEW

The module talks to other modules through events, not direct calls. Every event uses one envelope:

```
{ "eventId": "6705a1c2e4b0f1a2b3c4d6a0", "type": "vitals.alert.raised", "schemaVersion": 1,
  "occurredAt": "2026-10-09T15:03:02+05:30", "patientId": "6701f0a9e4b0f1a2b3c4d001",
  "requestId": "b1c2…", "data": { … } }
```

Event	Sent when	`data`	Consumer
`vitals.log.updated`	A save changes at least one value	`date`, `logId`, `changedTypes`, `filledCount`, `dayStatus`	Timeline ("Readings updated"), dashboard
`vitals.alert.raised`	An alert is created, or its severity goes up	`alertId`, `logId`, `type`, `field`, `value`, `rule`, `threshold`, `severity`, `doctorIds`	Notifications (notifies the linked doctors), Timeline
`vitals.alert.acknowledged`	A doctor acknowledges an alert	`alertId`, `acknowledgedBy`, `acknowledgedAt`	Timeline, dashboard

- **Delivery:** at least once. Events are written to the outbox in the save transaction and published only after it commits. Failed publishes retry with backoff (1 s, 5 s, 30 s, then every 5 min) for 24 hours, then go to a dead-letter list for review.
- **Consumers** must ignore an `eventId` they have already handled.
- **Ownership:** this module publishes. The Notifications module decides how doctors are notified (push, in-app).
- **Versions:** fields may be added within a `schemaVersion`. Removing or renaming a field raises it.

Audit log NEW

Each record holds `actorId`, `actorRole`, `patientId` (if any), `action`, `route`, `resourceId`, `before`, `after`, `requestId` and `at`. Records are never edited or deleted.

action	Recorded when	before / after
`VITALS_VALUE_CHANGED`	A patient changes or clears a value	Old and new value of each changed field
`PATIENT_VITALS_VIEWED`	A doctor calls a patient's today, logs or trends	None
`ALERT_ACKNOWLEDGED`	A doctor acknowledges an alert	Status, note
`THRESHOLD_CHANGED`	An admin sets an alert limit	Previous and new version
`VITAL_TYPE_CHANGED`	An admin changes a parameter	Previous and new version

Change records are written in the same transaction as the change.

Rate limits NEW

Scope	Limit
`PUT /me/vitals/today`	30 requests per minute per user
All other routes	120 requests per minute per user
Any route, by IP address	300 requests per minute
Size of one PUT	13 readings at most, request body up to 16 KB

A `429` response includes a `Retry-After` header in seconds. These are starting values, to be tuned after testing.

Errors CHANGED

Every error uses one format. `code` is stable, so the app can act on it. `message` and `issue` are for people and may change.

```
{ "statusCode": 400, "code": "VALIDATION_FAILED", "error": "Bad Request",
  "message": "Validation failed", "requestId": "b1c2…",
  "details": [ { "field": "readings.BP.systolic", "code": "BP_SYSTOLIC_NOT_ABOVE_DIASTOLIC",
                 "issue": "must be greater than diastolic" } ] }
```

Status	code	When
400	`VALIDATION_FAILED`	Body or query fails validation. `details[].code` is one of `REQUIRED`, `UNKNOWN_TYPE`, `WRONG_SHAPE`, `OUT_OF_RANGE`, `TOO_LONG`, `BP_SYSTOLIC_NOT_ABOVE_DIASTOLIC`, `EMPTY_READINGS`
401	`UNAUTHENTICATED`	Token missing, invalid or expired
403	`FORBIDDEN_ROLE`	Caller's role can't use this route
404	`NOT_FOUND`	Record not found, or belongs to a patient the caller can't access
409	`DAY_CHANGED`	The `date` sent is no longer the patient's today

Status	code	When
409	`ALERT_ALREADY_ACKNOWLEDGED`	Alert already acknowledged
409	`CONCURRENT_UPDATE`	A concurrent write changed the record first, so nothing was saved. The same request can be retried. Raised by a lost version check on `PUT /admin/vital-types/:type` and by a duplicate-key clash on a unique index
429	`RATE_LIMITED`	Rate limit exceeded
500	`INTERNAL`	Unexpected server error. Details are logged, never returned

> `CONCURRENT_UPDATE` was added on 14 Oct 2026, during implementation, and is not part of the set reviewed on 13 Oct. It covers a case the table had no honest code for: a lost race, where the record exists and nothing was saved. `404` would claim the record is missing, and `DAY_CHANGED` would send the app off to reload today's log as though midnight had passed.

Open questions

> All alert limits in this document (pain above 6, pulse 50–110) are placeholders until the clinical team confirms them. The system ships with no global limits until they are signed off. Until then, alerts come only from limits an admin enters.

Question	Owner
Alert limits and severity (low / medium / high) for each parameter, for example SpO2 below what %?	Clinical team
Confirm the data-entry valid ranges in the Parameters table	Clinical team
Format for nutrition intake: free text or fixed options? (Free text, up to 500 characters, until decided.)	Clinical team, frontend
Labels for mood levels 1, 2, 3 and 5 (level 4 is "Good")	Frontend team
Can patients correct a previous day's log? (Read-only until decided.)	Clinical team
Blood sugar is optional. Should a day be `COMPLETE` without it, for example 12 of 14?	Frontend team

Changes in this version

Version 1.1 answers the Health Monitoring API review of 13 Oct 2026.

Review item	Change
Duplicate writes and retries (P0)	Added *Duplicates and retries*: one value per parameter per day, unchanged values are skipped, so PUT is idempotent. Added a required `date` so a retry after midnight can't land on the next day.

Review item	Change
Edit / delete policy (P0)	Added *Reading lifecycle*: today editable, past days read-only, nothing deleted, every change audited.
13 parameter definitions (P0)	Valid range added for every parameter. Parameter versioning added. Remaining labels and formats are in Open questions.
Clinical thresholds (P0)	Still placeholders, now stated plainly with no global limits shipped until sign-off. BP limits are set per field. Thresholds are versioned.
Trend semantics, timezone (P0/P1)	Trend value per day defined for every parameter. Added *Day and time zone*. Thresholds in trends defined.
BP validation (P1)	Separate range per field, in `GET /vital-types` and thresholds.
Transaction and async (P1)	Added *Save flow* with an outbox: what is in the transaction and what runs after.
Events (P1)	Added *Events*: names, envelope, payloads, retries, consumer de-duplication.
Alert lifecycle (P1)	Added *Alert lifecycle*. Renamed `reviewedBy` / `reviewedAt` to `acknowledgedBy` / `acknowledgedAt`.
Audit (P1)	Added *Audit log* covering acknowledgements and admin changes.
Rate limits	Added limits and `Retry-After`.
Error codes (P1)	Added a stable `code` to every error and to each detail.
Full schemas	Added missing query tables, responses and errors for logs, trends, admin and acknowledgement.

The review also asked for the full endpoint list and data model in the Architecture Plan. That change belongs in the Architecture Plan, not in this reference.